

NGUYEN CAO XUAN PHUC

INNOVATION & TECHNOLOGY ENTHUSIAST | STUDENT RESEARCHER | SOCIAL
IMPACT LEADER

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PROFILE

I am a high school student interested in innovation, technology, and sustainable development. Through interdisciplinary projects in AI, environmental protection, assistive technology, and digital research, I have developed experience in teamwork, project coordination, communication, and problem-solving. I hope to combine technology with business and social responsibility to create practical solutions that contribute to communities and long-term sustainable development.

EDUCATION & ACADEMIC CREDENTIALS

Nguyen Hue High School for the Gifted

Hanoi, Vietnam

Specialized Physics Program | Grade 11

Academic record: Grade 10 GPA 9.1/10; Grade 11 GPA 9.3/10

SAT 1440(math 800/800) | IELTS 7.5

AP Physics C: 5/5

SELECTED HONORS & AWARDS

Gold Award - World Invention Creativity Olympic (WICO 2026)

Jul 2026 | Seoul, Korea

Project: SafeStride - A High-Precision Spatial Awareness and Priority Alert System for the Blind on Smartphones

- Developed an assistive smartphone concept designed to improve spatial awareness and deliver priority alerts for blind users.
- Awarded Gold; the certificate recognizes the team's presentation quality, creativity, and innovation.

First Prize, Arts & Technology Award, and Global Nominee - NASA

Oct 2025 | Ninh Binh

International Space Apps Challenge

Team BlueMarble | Project: BloomWatch

- Created an Earth-observation learning platform that transforms NASA satellite data on global flowering phenomena into an accessible multimedia experience.
- Integrated data research, AI-generated characters, visual storytelling, and collaborative product development during a 48-hour hackathon.

Third Prize - Environmental Protection Ideas in the Transport Sector

Jun 2026 | Hanoi

Competition

University of Transport and Communications | Project: ECOMeSort

- Designed an AI-enabled mobile application that guides waste sorting, connects users within the recycling ecosystem, and promotes environmental action through green-community features.

Member, NGS Hogrider

- Contributed to a team recognized for robot construction quality, reliability, craftsmanship, and competition readiness.

International Academic Awards**2021-2022**

- Bronze Award - Thailand International Mathematical Olympiad (TIMO), Heat Round 2021-2022.
- Bronze Award - Philippine International Mathematical Olympiad (PIMSO), Heat Round 2022.
- Merit Award, National Level - ASMO English 2022.

RESEARCH EXPERIENCE & SCHOLARLY ENGAGEMENT**Research Intern (Ongoing) - Institute of Information Technology,
VAST****18 May-24 Sep 2026 |
Hanoi**

AI Systems Department, Vietnam Academy of Science and Technology

- Participating in a six-month structured research internship at the AI Systems Department.
- Contributed to the project “A conversational robot companion for the elderly,” working on data labeling, interaction design, and system testing to help the robot understand and respond to older users more reliably.
- Gained hands-on experience with applied AI systems, scientific workflows, technical documentation, and interdisciplinary collaboration.

**Student Author and Presenter - VNICT 2025, 28th National
Conference on Selected ICT Problems****Nov 2025 | Ninh Binh**

- Co-authored and presented a conference paper on an improved dual-image reversible data-hiding method designed to increase embedding capacity for digital-content authentication.
- Developed early experience in technical writing, experimental reasoning, academic presentation, and peer-facing scientific communication.

**Student Co-Author and Presenter - National Conference on Flight Corridors and Low-Altitude
Airspace Management (UTM), Hanoi**

Paper: An AI-Integrated UAV–Ground Control Station Framework for Situational Awareness and Search-and-Rescue, toward UTM Deployment in Low-Altitude Airspace

- Co-authored the paper, took part in experiments, and presented the work at a national conference on unmanned traffic management for low-altitude airspace.
- Proposed an integrated UAV and ground-control-station framework using AI to improve situational awareness and support search-and-rescue operations.

Self-study: Completed and received certificates for the following Coursera courses:

Building on the GenAI training above, I pursued these courses on my own initiative to move from using AI tools toward understanding the systems behind them, in support of my interest in hardware and semiconductors.

- AI Fundamentals
- Vibe Coding with Claude Code
- AI for Data Analysis
- Vibe Coding with GitHub Copilot

LEADERSHIP, ENTREPRENEURSHIP & SOCIAL IMPACT

Founder & Project Chair - ECOMe	2025-Present
<i>Student-led environmental education and community-action initiative</i>	
– Founded and led awareness campaigns reaching approximately 1,800 people through field activities in the Mekong Delta and multiple Vietnamese provinces, including Hai Duong and Quang Ninh. – Led a June 2026 campaign that combined environmental communication with community service at six orphanages and child-care centers, supporting projects such as clean-water wells, rural bridges, street lighting, community kitchens, and livelihood assistance. – Initiated and chaired an environmental-ideas competition; coordinated communications, participant engagement, and project promotion. – Project evidence: ECOMe activity page	
Community Service Volunteer and Campaign Organizer	2025-2026
– Participated in charitable programs across Bac Ninh, Hai Duong, Lang Son, Dak Lak, the Mekong Delta, and other localities. – Supported two 2025 campaigns that delivered a combined VND 115 million in gifts and aid; also joined relief efforts serving 225 households affected by natural disasters in Dak Lak.	
Volunteer Recognitions - University of Transport and Communications	(2026 Hanoi)
– Certificate of Merit for outstanding contribution to community volunteer activities in 2026 (Aug 2026). – Certificate of Merit for outstanding contribution to the “Exam Support” and “Green Summer” volunteer campaigns in 2026 (Sep 2026). – Volunteer, “Green Sunday” community clean-up program 2026 (Mar 2026). – Volunteer and logistics/reception support for the 3rd Rectors’ Conference of Vietnam–Russia Universities and the Russian Federation Higher Education Fair (May 2026). – Co-organizer of the Youth Month launch event and a Generative AI (GenAI) training session for student-union officers 2026 (Mar 2026) — helping run this session first sparked my interest in AI tools and directly motivated my later self-study on Coursera.	
STEM PROGRAMS, INTERNATIONAL EXPOSURE & SCHOOL LEADERSHIP	
Open Data Exploration 2026 Summer Camp - Organizing Team Member	Jul 2026 American Center, U.S. Embassy Hanoi
– Served in two functions: Communication & Design and Training & Data Research, supporting program delivery, educational content, and participant engagement.	
Robotics Summer Camp 2026 - Mechanical Engineering Team Member	Jul 2026 American Center, U.S. Embassy Hanoi
– Supported free robotics education for Vietnamese students through technical teamwork and hands-on camp activities.	
AI for Good Vietnam 2026 - Team ECOMeSort	May 2026 Hanoi
– Participated in an education program focused on applying AI to socially beneficial challenges.	
Campus Tour and Learning Activities - University of Tsukuba	Jul 2025 Japan
– Completed academic and cultural learning activities hosted by the university's international student support office.	
Media & Community Roles	2024-2026
– Media Member, CNH Basketball Club - Panthers (2024-2025 and 2025-2026). – Media Department Collaborator, Net Dep Khoi Chuyen 2025. – Volunteer, Japanese Festival in Hanoi 2025.	

CORE COMPETENCIES & ACADEMIC INTERESTS

Research & Innovation

Scientific writing; reversible data hiding; digital-content authentication; problem formulation; evidence-based presentation.

Leadership & Communication

Project management; team collaboration; public outreach; media strategy; visual design; event organization.

Technology & Product Thinking

AI-enabled applications; assistive technology; robotics; data-driven storytelling; human-centered solution design.

Languages

Vietnamese (native); English (IELTS Academic 7.5).